

DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 5 (AT2)

Question 1 — Chart Selection

1. Comparing total revenue across 6 product categories in one year

Best chart type: Amounts → Bar Chart

Justification: A bar chart is ideal for comparing discrete categories because the lengths of the bars make differences in revenue easy to see and compare accurately.

2. Showing how the salary distribution differs across 4 job levels

Best chart type: Distribution → Box Plot

Justification: A box plot summarizes the median, quartiles, spread, and potential outliers, making it effective for comparing salary distributions across multiple job levels.

3. Showing market share of 5 competing brands

Best chart type: Proportion → Pie Chart

Justification: A pie chart is suitable when the goal is to show how each brand contributes to the whole market share and the categories are few in number.

4. Showing the relationship between advertising spend and sales revenue

Best chart type: x-y Relationship → Scatter Plot

Justification: A scatter plot displays paired numerical values and helps reveal correlation, trends, clusters, or outliers between advertising spend and sales revenue.

5. Showing monthly website traffic over 3 years

Best chart type: Temporal → Line Chart

Justification: A line chart is designed for time-series data and clearly shows trends, seasonality, and changes in website traffic over time.

6. Showing which employees report to which managers in an org

Best chart type: Network → Organizational / Hierarchical Network Diagram

Justification: Reporting relationships are hierarchical connections rather than numerical comparisons, so a network-style org chart best represents manager–employee structures.

Question 2 — Average Score Bar Plot

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt

# Dataset
df = pd.DataFrame({
    'Subject': ['Mathematics', 'Science', 'English', 'History', 'Computer Sci'],
    'Average_Score': [72, 68, 75, 64, 81]
})

# Sort descending
df = df.sort_values('Average_Score', ascending=False)

# Colors
colors = ['steelblue'] * len(df)
colors[0] = 'orange' # highest score

# Plot
plt.figure(figsize=(8,5))
bars = plt.bar(df['Subject'], df['Average_Score'], color=colors)
```

Labels

for bar in bars:

```
plt.text(bar.get_x() + bar.get_width()/2,  
         bar.get_height() + 1,  
         str(int(bar.get_height())),  
         ha='center')
```

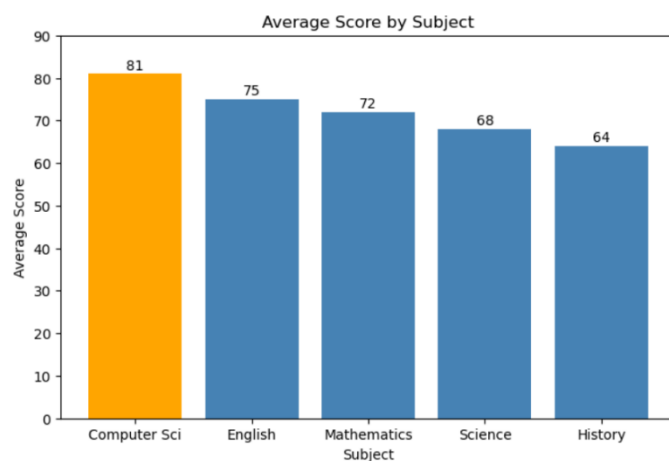
```
plt.title('Average Score by Subject')
```

```
plt.xlabel('Subject')
```

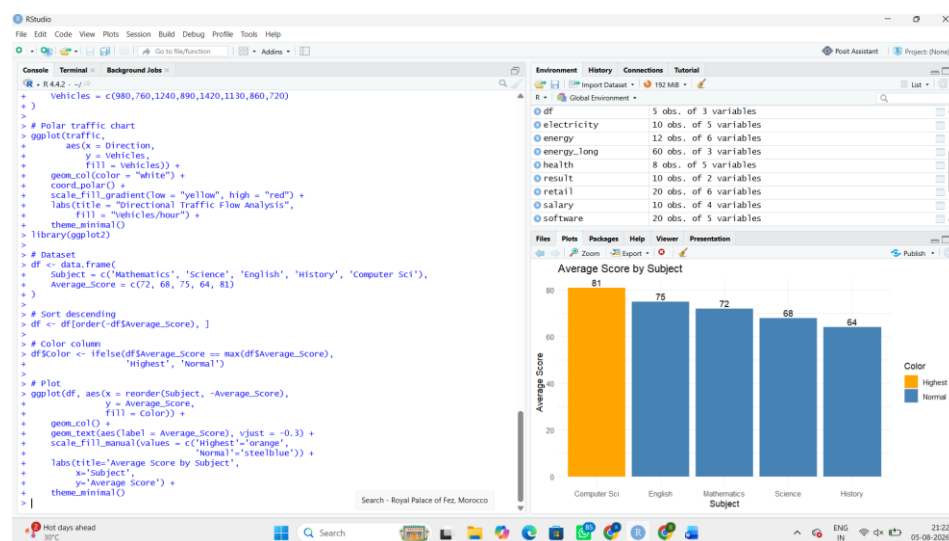
```
plt.ylabel('Average Score')
```

```
plt.ylim(0, 90)
```

```
plt.show()
```



R CODE



Question 3 — Regional Sales Visualizations

Python Code

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Dataset
```

```
df = pd.DataFrame({  
    'Region':['North','North','North','North',  
              'South','South','South','South',  
              'East','East','East','East',  
              'West','West','West','West'],  
    'Quarter':['Q1','Q2','Q3','Q4']*4,  
    'Sales_000s':[120,135,128,150,  
                  95,110,105,130,  
                  140,138,145,160,  
                  80,85,90,100]  
})
```

```
pivot = df.pivot(index='Region', columns='Quarter', values='Sales_000s')
```

```
# 1. Grouped Bar Chart
```

```
pivot.plot(kind='bar', figsize=(8,5))
```

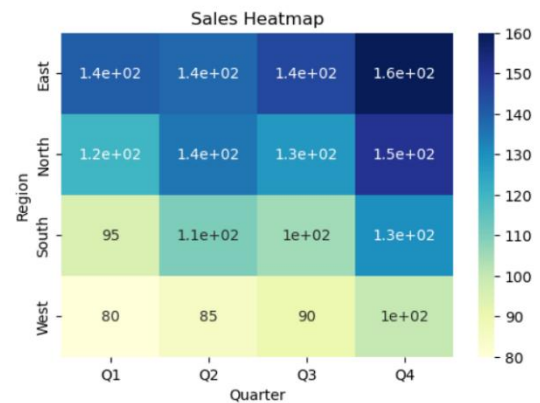
```
plt.title('Sales by Region and Quarter')
```

```
plt.xlabel('Region')
```

```
plt.ylabel('Sales (000s)')
```

```
plt.show()
```

```
# 2. Stacked Bar Chart
```



```
pivot.plot(kind='bar', stacked=True, figsize=(8,5))
```

```
plt.title('Stacked Quarterly Sales by Region')
```

```
plt.xlabel('Region')
```

```
plt.ylabel('Sales (000s)')
```

```
plt.show()
```

3. Dot Plot

```
totals = df.groupby('Region')['Sales_000s'].sum().sort_values(ascending=False)
```

```
plt.figure(figsize=(6,4))
```

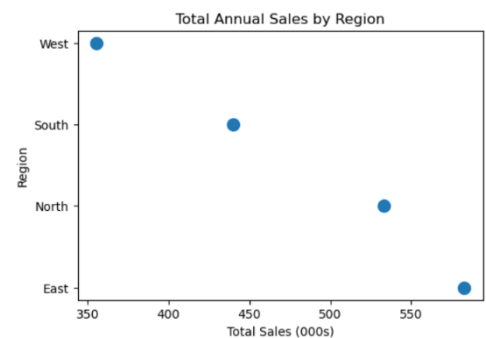
```
plt.scatter(totals.values, totals.index, s=100)
```

```
plt.title('Total Annual Sales by Region')
```

```
plt.xlabel('Total Sales (000s)')
```

```
plt.ylabel('Region')
```

```
plt.show()
```



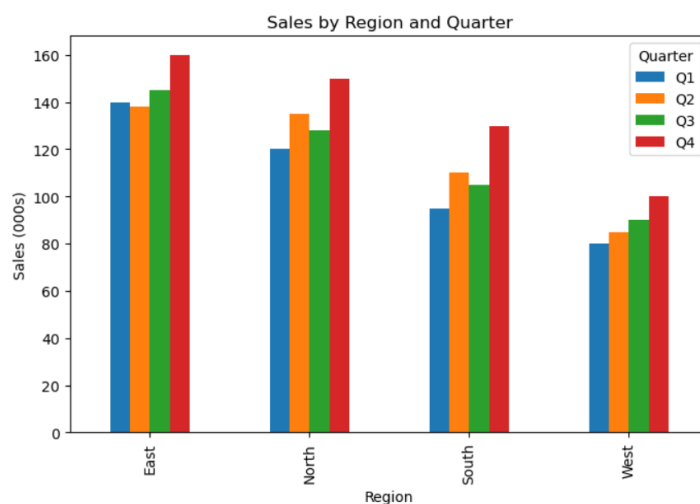
4. Heatmap

```
plt.figure(figsize=(6,4))
```

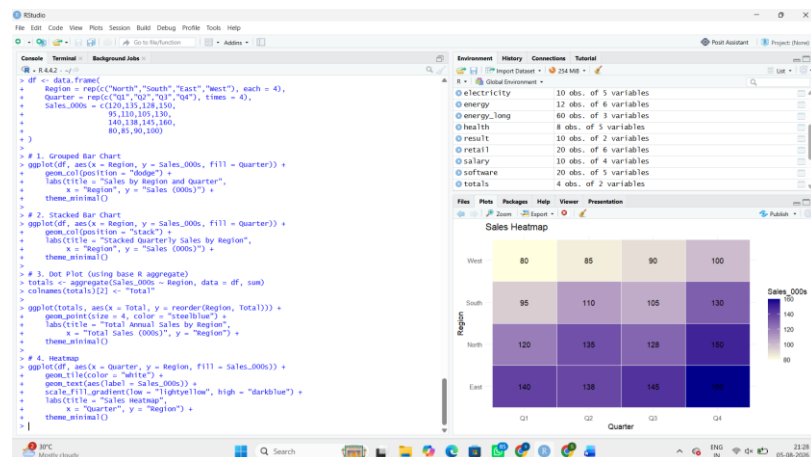
```
sns.heatmap(pivot, annot=True, cmap='YlGnBu')
```

```
plt.title('Sales Heatmap')
```

```
plt.show()
```



R CODE



Question 4 — Synthetic Exam Scores

Python Code

```
import numpy as np
```

```
import pandas as pd
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
# Reproducibility
```

```
np.random.seed(42)
```

```
# Generate synthetic data
```

```
data = pd.DataFrame({  
    'Score': np.concatenate([  
        np.random.normal(70, 8, 40),  
        np.random.normal(75, 5, 40),  
        np.random.normal(65, 12, 40),  
        np.random.normal(80, 6, 40)  
    ]),  
    'Class': ['A'] * 40 + ['B'] * 40 + ['C'] * 40 + ['D'] * 40  
})
```

```
# -----
```

```
# 1. Violin Plot
```

```
# -----
```

```
plt.figure(figsize=(8, 5))
```

```

sns.violinplot(
    data=data,
    x='Class',
    y='Score',
    inner='box',
    palette='Set2'
)

plt.title('Exam Score Distribution by Class')
plt.xlabel('Class')
plt.ylabel('Exam Score')
plt.show()

# -----

# 2. Ridgeline-style KDE Plot
# -----

# Order classes by mean score
order = (
    data.groupby('Class')['Score']
        .mean()
        .sort_values()
        .index
)

# Create overlapping KDE plots
plt.figure(figsize=(8, 5))

for cls in order:
    subset = data[data['Class'] == cls]
    sns.kdeplot(
        subset['Score'],
        fill=True,
        alpha=0.5,
        linewidth=1.5,
        label=f'Class {cls}'

```



```
)

plt.title('Ridgeline-style Density Plot of Exam Scores')

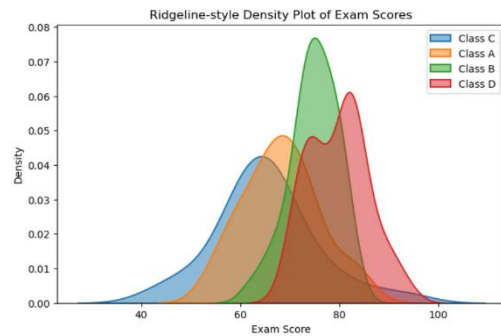
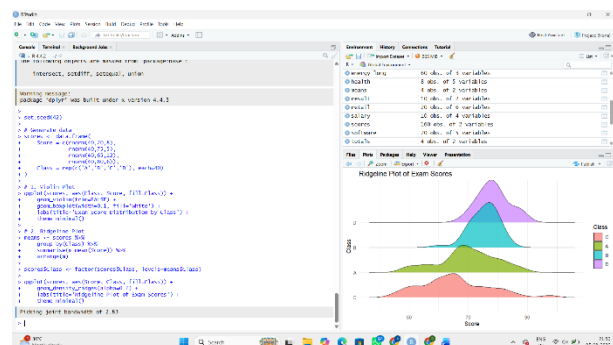
plt.xlabel('Exam Score')

plt.ylabel('Density')

plt.legend()

plt.show()
```

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Question 5 — Histogram + KDE

Python Code

```
import seaborn as sns

import matplotlib.pyplot as plt

tips = sns.load_dataset('tips')

# 1. Overall Histogram + KDE

plt.figure(figsize=(7,5))

sns.histplot(tips['total_bill'], bins=30, kde=True)

plt.title('Distribution of Total Bill')

plt.xlabel('Total Bill')

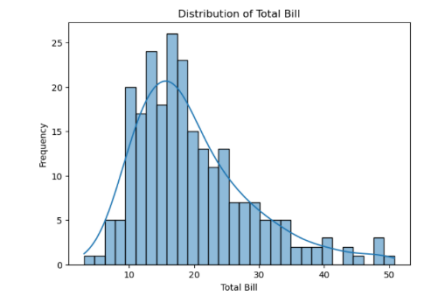
plt.ylabel('Frequency')

plt.show()

# 2. Grouped by Smoker

plt.figure(figsize=(7,5))

sns.histplot(data=tips,
```



```

x='total_bill',
hue='smoker',
bins=30,
kde=True,
element='step')

plt.title('Total Bill Distribution by Smoker')

plt.xlabel('Total Bill')

plt.ylabel('Frequency')

plt.show()

```

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